

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

5

Code of Conduct:

*I **B.Bhargavi 192311136** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **B.Bhargavi**

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Introduction

A healthcare company has collected patient information such as age, blood pressure, cholesterol, glucose level, BMI, and family history. The objective is to build a machine learning model that can predict whether a patient is diabetic or non-diabetic.

The company also wants to develop an AI agent that can recommend suitable treatment-related actions such as Diet, Exercise, Medication, or Monitor based on changes in the patient's health condition over time.

The complete system uses:

1. Data Preparation
2. Inductive Learning
3. Decision Tree Classification
4. Statistical Learning using Logistic Regression
5. Reinforcement Learning using Q-Learning

TASK 1: DATA PREPARATION AND INDUCTIVE LEARNING

1(a) Inductive Learning Approach

Meaning of Inductive Learning

Inductive learning is a machine learning approach in which the system learns general patterns from previously observed examples and uses those patterns to make predictions for new data.

In this healthcare problem, historical patient records are used for training. Each patient record contains several features and a known diabetes outcome.

For example:

A ge	Blood Press ure	Choleste rol	Gluc ose	B M I	Fami ly Hist ory	Diabe tes
25	120	180	90	22	No	No
4	145	220	180	31	Yes	Yes

Age	Blood Pressure	Cholesterol	Glucose	BMI	Family History	Diabetes
55	150	240	190	34	Yes	Yes
30	118	175	95	24	No	No

The machine learning algorithm learns the relationship between the patient features and the diabetes outcome.

Type of Learning

This problem uses **supervised inductive learning** because the training dataset contains labelled examples.

The model learns:

Patient Features → Diabetes Status

The target variable is:

Diabetes

It can be represented as:

- 0 = Non-Diabetic
- 1 = Diabetic

Important Features

1. Glucose Level

Glucose is one of the most important predictors of diabetes. Increased glucose levels can indicate abnormal blood sugar regulation.

Therefore, glucose is expected to have high predictive importance.

2. BMI

BMI indicates the relationship between body weight and height. A higher BMI is often associated with increased risk of type-2 diabetes.

Therefore, BMI can help the model identify patients with increased diabetes risk.

3. Age

Age can be an important risk factor because diabetes risk can change with increasing age.

For example, older patients may have a higher probability of developing type-2 diabetes compared with younger patients, although diabetes can occur at any age.

Other Important Features

Other useful features include:

- Blood pressure
- Cholesterol
- Family history

Family history can provide information about inherited or familial risk.

Therefore, the model can use all these features together to learn the diabetes prediction pattern.

1(b) Handling Class Imbalance

Meaning of Class Imbalance

Class imbalance occurs when the number of observations in one class is much greater than the number of observations in another class.

Here, non-diabetic patients form the majority class and diabetic patients form the minority class.

If the imbalance is ignored, the model may become biased toward predicting the majority class.

For example, if a model predicts almost every patient as non-diabetic, it may still obtain high accuracy because most patients are non-diabetic.

However, this is dangerous in healthcare because diabetic patients may be incorrectly classified as non-diabetic.

This type of error is called a **False Negative**.

Suitable Strategy: SMOTE

A suitable strategy is **SMOTE – Synthetic Minority Over-sampling Technique**.

SMOTE creates synthetic examples for the minority class instead of simply copying existing records.

For example, if there are:

- 800 non-diabetic patients
- 200 diabetic patients

SMOTE can generate additional synthetic diabetic samples to make the classes more balanced.

Advantages of SMOTE

1. It increases the number of minority-class examples.
2. It helps the model learn diabetic patterns better.
3. It reduces bias toward the majority class.
4. It can improve recall for diabetic patients.
5. It can reduce False Negatives.

Important Point

SMOTE should be applied **only to the training data**, not to the test data.

Correct process:

Original Dataset → 70:30 Split → Apply SMOTE to Training Data → Train Model → Test on Original Test Data.

This prevents data leakage.

Alternative: Class Weights

Another suitable technique is class weighting.

The model can assign a higher penalty to misclassifying diabetic patients.

This is useful because it directly tells the model that incorrectly classifying a diabetic patient is more costly.

Under-Sampling

Under-sampling removes some records from the majority class.

For example:

800 Non-Diabetic → 300 Non-Diabetic

and:

200 Diabetic → 200 Diabetic

The advantage is that the dataset becomes smaller and balanced.

However, some useful majority-class information may be lost.

Final Choice

For this problem, **SMOTE is a suitable balancing strategy** because the diabetic class is smaller and it allows the model to learn more minority-class patterns.

However, **class weighting is also highly suitable for a clinical system**, because it avoids creating synthetic patient records.

1(c) 70:30 Data Split

The dataset should be divided into:

- 70% Training Data
- 30% Testing Data

For example, if the dataset contains 1,000 patient records:

- 700 records are used for training.
- 300 records are used for testing.

Why 70:30?

The training dataset requires enough records for the model to learn patterns.

The testing dataset must contain enough unseen records to evaluate how well the model performs on new patients.

For a medical application, a **stratified split** is preferred.

Stratification maintains approximately the same diabetic and non-diabetic proportions in both training and testing datasets.

For example:

Original:

- 80% Non-Diabetic
- 20% Diabetic

Training:

- Approximately 80% Non-Diabetic
- Approximately 20% Diabetic

Testing:

- Approximately 80% Non-Diabetic

- Approximately 20% Diabetic

Preprocessing Steps

1. Missing Value Handling

Missing numerical values can be replaced using median values.

For example, if some patients have missing BMI values, the median BMI can be used.

For categorical features, the most frequent category can be used.

2. Removing Duplicate Records

Duplicate patient records should be identified and removed when appropriate.

This prevents the same patient information from influencing the model multiple times.

3. Encoding

Categorical variables must be converted into numerical values.

For example:

Family History:

- No = 0
- Yes = 1

If there are multiple categories, one-hot encoding can be used.

4. Normalisation

Features can have different numerical ranges.

For example:

- Age: 18–80
- Glucose: 50–300
- BMI: 15–50

Min-Max normalization can be used:

This converts values to approximately the range 0 to 1.

Impact of Normalisation

Normalization:

- Places numerical features on comparable scales.
- Improves numerical stability.

- Is useful for Logistic Regression.
- Prevents large-scale variables from dominating scale-sensitive algorithms.

Decision Trees generally do not require normalization because they split data using thresholds.

TASK 2: DECISION TREE FOR DIAGNOSIS

2(a) Building a Decision Tree

A Decision Tree is a supervised classification algorithm that makes decisions using a sequence of conditions.

For example, the model may first check glucose level.

If glucose is high, it may then check BMI.

If BMI is high, it may predict a higher diabetes risk.

The tree contains:

- Root Node
- Internal Nodes
- Branches
- Leaf Nodes

Root Node Selection

The root node is selected using measures such as:

1. Information Gain
2. Gini Index

The feature with the highest Information Gain or lowest Gini impurity is generally selected.

Information Gain

Information Gain measures how much uncertainty is reduced after splitting the data.

Entropy is:

Information Gain is:

A higher Information Gain indicates a better split.

2(b) Evaluation of the Decision Tree

The model can be evaluated using a confusion matrix.

Actual / Predicted	Non-Diabetic	Diabetic
Non-Diabetic	TN	FP
Diabetic	FN	TP

Where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

Accuracy

Accuracy represents the percentage of correctly classified patients.

Precision

Precision indicates how many patients predicted as diabetic are actually diabetic.

Recall

Recall indicates how many actual diabetic patients were correctly detected.

F1-Score

F1-score balances precision and recall.

2(c) Post-Pruning

A Decision Tree can grow very deep and memorize training data.

This is called overfitting.

For example:

Training accuracy = 99%

Testing accuracy = 75%

This indicates that the model performs well on training data but poorly on unseen patients.

Cost-Complexity Pruning

Cost-complexity pruning reduces unnecessary branches.

The cost function is:

where:

- E = error of the tree
- N = number of leaf nodes
- α = complexity parameter

A larger α results in more pruning.

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

3(a) Logistic Regression

Logistic Regression is suitable because diabetes classification is a binary classification problem.

The target is:

- 0 = Non-Diabetic
- 1 = Diabetic

The model estimates the probability that a patient is diabetic.

The equation is:

The logistic function is:

The result is a probability between 0 and 1.

For example:

Patient A $\rightarrow$ Probability = 0.82

This means the model estimates an 82% probability of the positive class, subject to the model's calibration and assumptions.

Comparison with Decision Tree

Example:

Metric	Decision Tree	Logistic Regression
Accuracy	86%	84%
AUC-ROC	0.89	0.88

Metric	Decision Tree	Logistic Regression
Interpretability	High	High
Non-linear patterns	Good	Limited
Probability estimation	Moderate	Good

The values are illustrative.

AUC-ROC

AUC stands for **Area Under the Receiver Operating Characteristic Curve**.

It measures how well the model separates diabetic and non-diabetic patients across different classification thresholds.

General interpretation:

AUC	Performance
0.50	Random
0.60–0.70	Poor
0.70–0.80	Acceptable
0.80–0.90	Good
Above 0.90	Excellent

A higher AUC indicates better discrimination.

When is Logistic Regression Preferable?

Logistic Regression is preferable when:

1. A probability or risk score is required.
2. A simple statistical model is required.
3. Interpretability is important.
4. Relationships between variables and outcome are reasonably well represented by the model.
5. Clinicians need to understand the direction and strength of predictor effects.

For example:

Patient A → 80% estimated risk

Patient B → 25% estimated risk

This type of probability output can be useful for risk stratification.

3(b) Feature Importance Analysis

Feature importance identifies the most influential variables in prediction.

Decision Tree

Suppose the feature importance is:

Feature	Importance	Rank
Glucose	0.45	1
BMI	0.20	2
Age	0.13	3
Blood Pressure	0.12	4
Cholesterol	0.06	5
Family History	0.04	6

The top three are:

1. Glucose
2. BMI
3. Age

Logistic Regression

For Logistic Regression, standardized coefficients or odds ratios can be examined.

Suppose the results show:

1. Glucose
2. BMI
3. Family History

The two models agree on glucose and BMI.

The third predictor differs.

Why do they differ?

Decision Trees can capture:

- Non-linear relationships
- Threshold effects
- Feature interactions

Logistic Regression models the relationship between predictors and the log-odds of the outcome.

Therefore, the two algorithms may rank features differently.

If both models consistently identify glucose and BMI as highly important, it provides useful evidence that these variables are strongly associated with prediction in the dataset.

However, feature importance should not be interpreted as proof that a feature causes diabetes.

TASK 4: REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

4(a) MDP Model

The treatment recommendation problem can be represented as a **Markov Decision Process**.

An MDP consists of:

where:

- S = States
- A = Actions
- P = Transition probabilities
- R = Reward
- γ = Discount factor

1. States

States represent the patient's health condition.

Example:

State	Patient Condition
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State	Patient Condition
S0	Normal / Low Risk
S1	Mild Risk
S2	High Risk
S3	Poorly Controlled
S4	Controlled / Improved

The state can be calculated from measurements such as:

- Glucose
- BMI
- Blood pressure
- Cholesterol
- Diabetes risk

For example:

High glucose + high BMI + high blood pressure → High-risk state.

2. Actions

The actions are:

1. Diet
2. Exercise
3. Medication
4. Monitor

Diet

The agent can recommend a suitable dietary intervention under clinical guidance.

Exercise

The agent can recommend an appropriate physical activity plan.

Medication

Medication is a high-risk action and should only be considered within a clinician-approved treatment protocol.

Monitor

The system continues monitoring the patient's health condition.

4(b) Q-Learning

Q-Learning is a Reinforcement Learning algorithm that learns the value of taking an action in a particular state.

It learns a Q-value:

The Q-learning update equation is:

where:

- α = Learning rate
- γ = Discount factor
- R = Reward
- s' = Next state

Assume:

and:

Initially all Q-values are 0.

Iteration 1

Suppose:

Current State = S2

Action = Diet

Reward = +5

Next State = S1

Initially, all Q-values for S1 are zero.

Therefore:

Therefore:

$$Q(S2, \text{Diet}) = 2.5$$

Iteration 2

Suppose:

Current State = S2

Action = Diet

Reward = +5

Next State = S1

Now assume:

Then:

Therefore:

$Q(S2, \text{Diet}) = 5.1$

The Q-value increased because the agent learned that Diet can lead to a better future health state.

ETHICAL CONSIDERATIONS

1. Patient Safety

Patient safety is the most important consideration.

An RL agent should not directly experiment on patients to learn which treatment works.

Unsafe actions must be restricted using clinical rules and safety constraints.

2. Bias

If the training dataset does not represent all relevant patient groups fairly, the model may perform differently across groups.

Therefore, the system should be evaluated for bias and fairness.

The company should check:

- Accuracy across groups
- Recall across groups
- False Negative rates
- False Positive rates

3. Explainability

Healthcare professionals need to understand why a recommendation was generated.

For example:

Patient State: High Risk

Important Factors: High glucose and high BMI

Suggested Action: Diet

Reason: The learned policy associates the current state and action with improved future health outcomes, subject to clinical constraints.

This makes the system easier to understand and audit.

4. Human Oversight

The AI should support doctors rather than replace them.

The recommended workflow is:

Patient Data → AI Prediction → Treatment Recommendation → Doctor Review → Final Decision

The doctor should have the authority to accept, reject, or modify the recommendation.

5. Privacy

Patient data is sensitive.

The system should use:

- Secure storage
- Access control
- Encryption
- Data de-identification where appropriate
- Proper data governance

6. Accountability

The system should maintain records of:

- Patient state
- Model version
- Recommended action
- Important features

- Clinician decision
- Patient outcome

This makes it possible to audit the system if an error occurs.

7. Continuous Monitoring

The AI model should be continuously monitored after deployment.

Its performance may change because patient populations, medical practices, and available treatments can change over time.

Therefore, periodic validation and model updating are required.

OVERALL CONCLUSION

The proposed system combines **supervised learning, Decision Tree classification, statistical learning, and Reinforcement Learning** to support diabetes diagnosis and treatment planning.

In Task 1, supervised inductive learning is used to learn the relationship between patient characteristics and diabetes status. Glucose, BMI, and age are important features, while blood pressure, cholesterol, and family history can also contribute to prediction. Class imbalance can be handled using SMOTE or class weighting, with SMOTE applied only to the training data.

In Task 2, the Decision Tree uses Information Gain or Gini Index to select the best splitting feature. Glucose may become the root node because it can provide strong separation between diabetic and non-diabetic patients. Accuracy, Precision, Recall, and F1-score are used for evaluation. Recall is particularly important because False Negatives represent diabetic patients who are incorrectly classified as non-diabetic. Post-pruning can reduce overfitting and produce a simpler model.

In Task 3, Logistic Regression provides a statistical approach for diabetes risk prediction. It can provide probability-based risk estimates and can be compared with the Decision Tree using Accuracy and AUC-ROC. Feature importance analysis helps identify the strongest predictors.

In Task 4, treatment recommendation is represented as an MDP consisting of states, actions, rewards, and transition dynamics. Q-Learning updates the Q-table based on patient outcomes and gradually learns the best action for each health state.

